

Lesson 5. Systems of Differential Equations II

University of Budapest

Faculty of Informatics, Department of Numerical Analysis

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Table of Contents

- 1 5.1. Linearization and Stability Analysis of Nonlinear Systems
- 2 5.2. Limit Cycles. Existence and Non-existence Criteria

Nonlinear Systems of ODEs

The general form of the nonlinear system of differential equations is

$$\frac{dx_i}{dt} = f_i(x_1, x_2, \dots, x_n), \quad i = 1, 2, \dots, n. \quad (1)$$

In vector form it can be written as

$$\mathbf{X}' = \mathbf{f}(\mathbf{X}), \quad \text{where } \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}.$$

Unlike linear systems, nonlinear systems can exhibit a wide range of complex behaviors:

- multiple equilibria;
- limit cycles;
- chaos;
- bifurcations.

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Linearization around Critical Points

To begin, we always start by finding the critical points, which correspond to the equilibrium solutions of the system. In the case of linear systems, the only critical point is the origin, $(0,0)$. Nonlinear systems can have many fixed points, and we want to determine the behavior of the trajectories near these points.

A critical point is **isolated** if it is the only critical point in some small "neighborhood" of the point. That is, if we zoom in far enough it is the only critical point we see. If on the other hand there would be a whole curve of critical points, then it would not be isolated.

The goal of linearization is to use our knowledge of linear systems to conclude what we can about the phase portrait near (x_0, y_0) . We will try to approximate our nonlinear system by a linear system, which we can then classify. Since (x_0, y_0) is a fixed point, and the only fixed point of a linear system is the origin, we will want to change variables so that (x_0, y_0) becomes the origin of the new coordinate system.

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Linearization around Critical Points

Theorem: Let $\mathbf{x}_0$ be a critical point for the system

$$\mathbf{x}' = \mathbf{F}(\mathbf{x}),$$

where the functions $f_i(\mathbf{X})$ are twice continuously differentiable in a neighborhood of the $\mathbf{x}_0$.

Then the system can be written as

$$\mathbf{x}' = (\mathbf{A} + \mathbf{E}(\mathbf{x}))[\mathbf{x} - \mathbf{x}_0], \quad (2)$$

where $\mathbf{A} = J(\mathbf{x}_0)$ is the Jacobian matrix of $\mathbf{F}$ at $\mathbf{x}_0$:

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \vdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \vdots & \frac{\partial f_2}{\partial x_n} \\ \dots & \dots & \vdots & \dots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \vdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}.$$

and $\mathbf{E}$ is a continuous matrix-valued function of position satisfying $\mathbf{E}(\mathbf{x}) \rightarrow \mathbf{0}$ as $\mathbf{x} \rightarrow \mathbf{x}_0$.

Linearization around Critical Points

Dropping $\mathbf{E}(\mathbf{x})$ in equation (2) gives us the shifted linear system $\mathbf{x}' = \mathbf{A}[\mathbf{x} - \mathbf{x}_0]$, often referred to as the linearization of the system about critical point $\mathbf{x}_0$.

Thus, in the study of the stability, we consider the linearized system

$$\mathbf{X}' = J\mathbf{X},$$

instead of the original nonlinear.

Is it really safe to neglect the term $\mathbf{E}(\mathbf{x})$?

In other words, does the linearized system give a qualitatively correct picture of the phase portrait near critical points $\mathbf{x}_0$?

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Stability of the linearized systems

Lyapunov theorem on stability in linearized systems

If all eigenvalues λ_i of the Jacobian J have **negative real parts**, then the zero solution of the original and linearized systems is **asymptotically stable**.

Lyapunov theorem on instability in linearized systems

If at least one of the eigenvalues λ_i of J has a **positive real part**, then the zero solution of the original and linearized systems is **unstable**.

In **critical cases**, when the real parts of all eigenvalues λ_i of the Jacobian J are non-positive and there is at least one eigenvalue with zero real part, the zero solution can be stable or unstable. In this case, it's impossible to determine the character of stability in the first approximation and one should use other methods of stability analysis.

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Stability of the linearized systems: Example

Example 1

Find all the fixed points of the system

$$\begin{cases} x' = -x + x^3, \\ y' = -2y, \end{cases}$$

and use linearization to classify them. Then check your conclusions by deriving the phase portrait for the full nonlinear system.

Solution:

Fixed points occur where $x' = 0$ and $y' = 0$ simultaneously. Hence, we need

$$\begin{cases} -x + x^3 = 0 \\ -2y = 0 \end{cases}$$

The second equation gives $y = 0$. The first equation gives $x = 0$ or $x = \pm 1$. Thus, there are three critical points: $(0, 0)$, $(1, 0)$, and $(-1, 0)$.

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The Jacobian matrix at a general point (x, y) is

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} -1 + 3x^2 & 0 \\ 0 & -2 \end{pmatrix}.$$

Next, we evaluate J at the fixed points.

$$J(0, 0) = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix},$$

so $(0, 0)$ is a **stable node**.

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At $(\pm 1, 0)$, we have

$$J(\pm 1, 0) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix},$$

so both $(1, 0)$ and $(-1, 0)$ are **saddle points**.

Now, because stable nodes and saddle points are not critical cases, we can be certain that the fixed points for the full nonlinear system have been predicted correctly.

As a final observation, we note that the phase portrait must be symmetric in both the x and y axes, since the equations are invariant under the transformations $x \mapsto -x$ and $y \mapsto -y$. Putting all this information together, we arrive at the phase portrait shown in the next slide.

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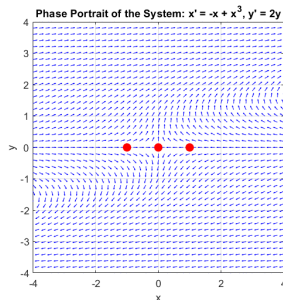
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Stability of the linearized systems: Example



The critical cases (centers, dicritical nodes, singular nodes, or non-isolated critical points) are much more delicate. They can be altered by small nonlinear terms.

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Periodic Solutions and Closed Orbits

A solution of $x' = f(x, t)$ through x_0 is said to be **periodic** if there exists a $T > 0$ such that $x(t + T, x_0) = x(t, x_0)$ for all $t \in \mathbb{R}$.

A solution of $x' = f(x, t)$ is said to be a **closed orbit** if the trajectory of the solution forms a closed curve in the phase space.

Closed Orbit

A closed orbit is thus a geometric representation of a periodic solution in the phase space, where the trajectory of the system repeats itself and traces a closed path.

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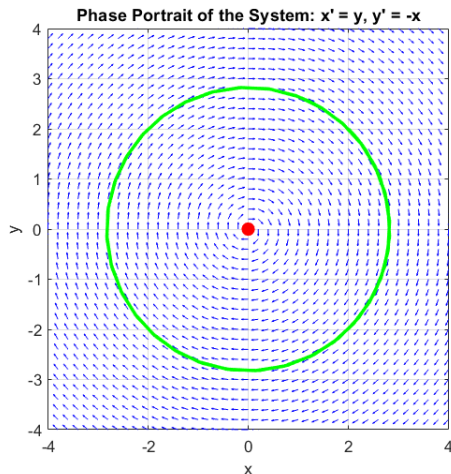
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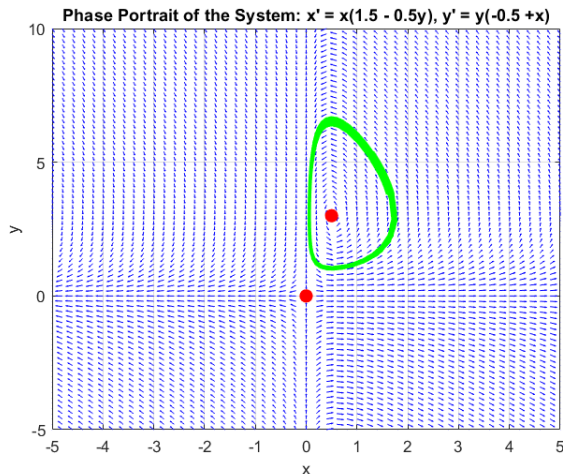
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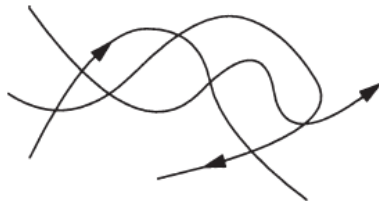
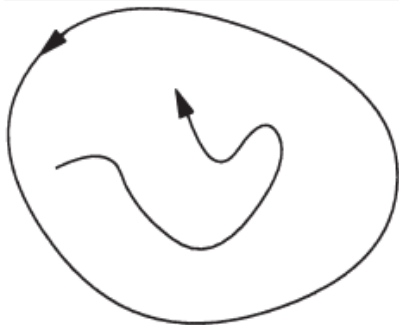


Example 2



Limit Cycles: Motivation

The existence and uniqueness theorem ensures that different trajectories in a phase portrait never intersect. If they did, it would mean two solutions start from the same point, violating the uniqueness part of the theorem.

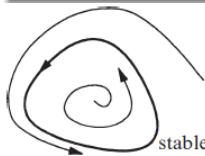


This non-intersecting behavior gives phase portraits their structured, organized appearance. In two-dimensional spaces, the theorem has significant topological implications: for example, if there is a closed orbit, any trajectory inside it remains trapped there.

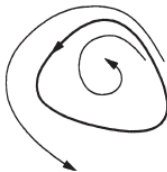
Limit Cycles: Definitions

A **limit cycle** is an isolated closed trajectory.

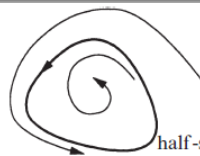
Isolated means that neighboring trajectories are not closed; they spiral either toward or away from the limit cycle.



stable
limit cycle



unstable
limit cycle



half-stable
limit cycle

If all neighboring trajectories approach the limit cycle, we say the limit cycle is **stable**.

Otherwise the limit cycle is **unstable**, or in exceptional cases, half-stable.

Limit Cycles: Definitions

- Example 1 : the equilibrium point $(0, 0)$ is **not** a limit cycle;
- Example 2 : the equilibrium point $(0.5, 3)$ is a limit cycle

From the perspective of qualitative analysis, our task is to find methods for determining the existence of closed orbits in specific systems. The following theorem is one of the few results that contribute to this objective.

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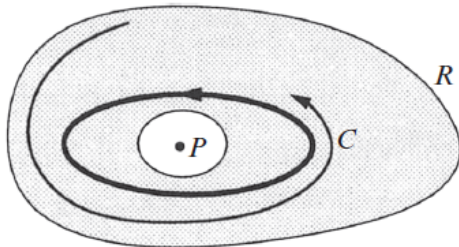
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Poincare-Bendixson Theorem

Poincare-Bendixson Theorem: Suppose that:

- ① R is a closed, bounded subset of the plane;
- ② $\dot{x} = f(x)$ is a continuously differentiable vector field on an open set containing R ;
- ③ R does not contain any equilibrium points; and
- ④ There exists a trajectory C that is "confined" in R , in the sense that it starts in R and stays in R for all future time

Then either C is a closed orbit, or it spirals toward a closed orbit as $t \rightarrow \infty$. In either case, R contains a closed orbit.



Dulac's Criteria and Bendixon's Criteria

Dulac's Criteria: Consider the autonomous planar system

$$\begin{aligned}x' &= f(x, y), \\y' &= g(x, y)\end{aligned}$$

and a continuously differentiable function ψ defined on a domain R contained in some open set. If

$$\frac{\partial}{\partial x}(\psi f) + \frac{\partial}{\partial y}(\psi g)$$

does not change sign in R , then there is at most one limit cycle contained entirely in R .

Bendixon's Criteria: Consider the autonomous planar system

$$x' = f(x, y), \quad y' = g(x, y)$$

defined on a simply connected domain R such that

$$\frac{\partial}{\partial x}(\psi f) + \frac{\partial}{\partial y}(\psi g) \neq 0$$

in R . Then there are no limit cycles entirely in R .

Dulac's Criteria and Bendixon's Criteria

Example 1

Consider the following system:

$$\begin{aligned}x' &= x - y - x(x^2 + y^2), \\y' &= x + y - y(x^2 + y^2)\end{aligned}$$

A simple computation gives that

$$\nabla \cdot \mathbf{f} = 2 - 4x^2 - 4y^2$$

For an arbitrary annulus $a < x^2 + y^2 < b$, we have

$$2 - 4b < \nabla \cdot \mathbf{f} < 2 - 4a$$

For $a = \frac{3}{4}$ and $b = \frac{5}{4}$, $-3 < \nabla \cdot \mathbf{f} < -1$. Thus, $\nabla \cdot \mathbf{f} < 0$ in the annulus $\frac{3}{4} < x^2 + y^2 < \frac{5}{4}$. Therefore, by Dulac's Criteria there is at most one limit cycle in this annulus.

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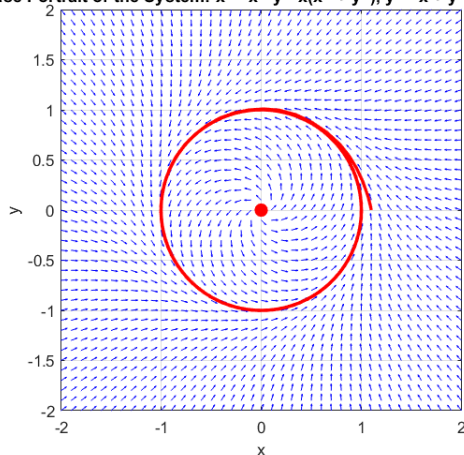
For an arbitrary annulus $a < x^2 + y^2 < b$, we have

$$2 - 4b < \nabla \cdot \mathbf{f} < 2 - 4a$$

For $a = \frac{3}{4}$ and $b = \frac{5}{4}$, $-3 < \nabla \cdot \mathbf{f} < -1$. Thus, $\nabla \cdot \mathbf{f} < 0$ in the annulus $\frac{3}{4} < x^2 + y^2 < \frac{5}{4}$. Therefore, by Dulac's Criteria there is at most one limit cycle in this annulus.

Example 1

Phase Portrait of the System: $x' = x - y - x(x^2 + y^2)$, $y' = x + y - y(x^2 + y^2)$



Dulac's Criteria and Bendixon's Criteria

Example 2

Consider the system

$$x' = y, \quad y' = -ax - by + cx^2 + dy^2.$$

Let $\psi(x, y) = e^{-2dx}$. Then,

$$\frac{\partial}{\partial x}(\psi y) + \frac{\partial}{\partial y}(\psi(-ax - by + cx^2 + dy^2)) = -be^{-2dx} \neq 0.$$

We conclude by Bendixon's Criteria that there are no limit cycles for this system.

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